Robbie Palmer

Principal Software Engineer & Engineering Manager

robbiepalmer.me GitHub LinkedIn

EXPERIENCE

Principal Software Engineer & Engineering Manager | Terminal Industries

May 2024 - Present

Building real-time computer vision products and ML pipelines, while driving engineering practices for a startup transitioning from pre-product to production.

- Built real-time computer vision products for multi-stream video and image analysis, for identifier identification in complex environments
- Engineered ML pipelines with data version control, benchmarking infrastructure, and end-to-end observability powered by Grafana Cloud (Prometheus, Loki, Tempo)
- Developed web services, Terraform managed infrastructure, and LLM-powered document analysis solutions

Technologies: Python, Terraform, Grafana, Prometheus, DVC, Google Gemini, TypeScript, GitHub Actions, TensorRT

Machine Learning Lead | Bestomer

May 2023 - Feb 2024

Led ML strategy at a seed-stage startup, building pipelines, evaluating build-vs-buy decisions, and developing full-stack prototypes.

- Led ML strategy and built end-to-end ML pipelines for training, evaluation, and deployment
- Optimized, evaluated, and integrated ML models powering semantic search, image similarity search, invoice extraction, product detection, and RAG-based chatbots
- Developed backend services in Python and TypeScript and contributed to iOS application development

Technologies: TypeScript, Node.js, Fastify, Prisma, Postgresql, Weaviate, Python, FastAPI, Pulumi

Customer Innovation Engineer II | Confluent

Apr 2022 - May 2023

Designed and built ML-driven data streaming solutions, wearing multiple hats across engineering, product, and customer-facing roles.

- Designed and engineered machine-learning-driven data-streaming solutions
- Acted as software engineer, product owner, project manager and technical consultant
- Built high-performance stream-processing applications and extensions for ksqlDB and Kafka Connect

Technologies: Kafka, Kotlin, Python, Google BigQuery, spaCy, SemaphoreCI, DVC

Senior Machine Learning Engineer | Sonrai Analytics

Mar 2020 - Apr 2022

Technical lead for Data Science at an early-stage health-tech startup, shaping engineering practices and building computer vision pipelines for pathology.

- Acted as technical lead for the Data Science team, making foundational architectural decisions and mentoring junior engineers
- Developed computer vision models for semantic segmentation, multiple instance learning and object localisation
- Engineered full machine learning pipelines, managing algorithms, data sampling, modelling, versioning and iteration

Technologies: Python, PyTorch, FastAPI, Plotly, Leaflet, Docker, AWS, Bitbucket Pipelines, DVC

TECHNICAL STRENGTHS

Python | DVC | AWS | Docker | FastAPI | GitHub Actions | Google Gemini | Leaflet | OpenRouter | TypeScript | Ansible | Bitbucket Pipelines | Cloudflare Workers | Fastify | GeoPandas | Go

EARLIER EXPERIENCE

Senior Software Engineer | Philips

Jul 2017 - Mar 2020

Led development of production-grade computer vision for computational pathology, from algorithm design through to scalable deployment.

- Led development of production-grade computer vision solutions for computational pathology
- Drove algorithm conception, data collection and engineering, and deep learning model development

Technologies: Python, OpenCV, Keras, Docker, Jenkins, RabbitMQ, Ansible

Research Assistant / Software Engineer Summer Intern | Ulster University

Jul 2016 - Sep 2016

Built Arduino-based CubeSat prototypes implementing university research on self-managing autonomous systems.

- Worked on SPAAACE project developing self-managing systems for CubeSats
- Built Arduino-based prototypes implementing autonomic computing principles

Technologies: Arduino, C++

Software Development Intern | Microsoft

Jul 2015 - Jun 2016

Improved Bing auto-suggest across all platforms through code optimization, big data analysis, and interactive dashboards.

- Designed, developed and tested C# code improving auto-suggest across all platforms
- Optimised large portions of the Bing search stack for performance

Technologies: C#, .NET, T-SQL

Software Development Intern | Infosys

Jun 2014 - Sep 2014

Built an NLP-powered system that generated graph databases from semantic analysis of web content.

- Developed a system integrating NLP analysis with graph database storage
- Built semantic relationship mapping for web-page content analysis

Technologies: Java, Stanford NLP, Neo4j

SELECTED PROJECTS

Work Graph

Source

A public source of truth for outcomes, dependencies, evidence, leases, and the human decisions that keep work moving

Cloudflare Access | Cloudflare Queues | Drizzle | DuckDB | dbt | GitHub | Hono | Hyperdrive

Personal Finance App

Source

Data-entry-first financial planning, the missing tool between budgeting apps and advisor software

Recipe Site

Source

A social cooking network that helps friends and families cook better using trusted, real-world kitchen signals

Autonomic Satellite Swarm

Source

A revived C++ research prototype for fail-safe coordination across hardware-independent satellite-swarm agents

C++ | CMake | Arduino | ESP32 | Catch2 | CesiumJS | Emscripten | WebAssembly

Agentic Code Review

Source

A stateful, model-agnostic GitHub App for high-signal code review at agentic development velocity

GitHub | Cloudflare Workers | Cloudflare Workflows | Cloudflare R2 | OpenRouter